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BSPD Design Documentation
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1. **Circuit Purpose & Safety Requirements**

The Brake System Plausibility Device (BSPD) is a standalone, non-programmable analog safety circuit used in the Formula Slug car. Powered from the **+5 V** supply rail, the circuit monitors hydraulic brake line pressure and tractive system DC current, opening the normally-closed Shutdown Circuit relay whenever an unsafe driving state or sensor wiring fault occurs:

Simultaneous Brake & Power Plausibility Fault: Triggers when hard mechanical braking occurs simultaneously with high tractive system power delivery (**≥ 5 kW** DC motor power).

Sensor Wire Integrity Faults (Short & Open Detection): Triggers immediately if either sensor output is pulled above **4.80 V** (short-circuit to +5 V power) or drops below **0.20 V** (broken wire or short-circuit to ground).

A. Reference Voltage Divider Equation (+5 V Supply Rail):

2. **Circuit Purpose & Safety Requirements**

$$V_{ref} = 5.0\ V \times [\ R_{bottom} \div (R_{top} + R_{bottom})\ ]$$

B. AiM 0–2000 PSI Brake Pressure Sensor (Linear 0.5 V to 4.5 V Output):

$$\text{Sensitivity} = (4.5\ V - 0.5\ V) \div 2000\ PSI = 0.002\ V/PSI \rightarrow V_{brake} = 0.5\ V + (0.002\ V/PSI \times PSI)$$

Selecting a **1000 PSI** threshold yields:
$\mathbf{V_{ref,brake} = 0.5\ V + (0.002 \times 1000) = 2.50\ V}$,
so $\mathbf{R5 = 10\ k\Omega}$ and $\mathbf{R6 = 10\ k\Omega}$. (For a 500 PSI threshold, V_{ref} = 1.50 V using 35 kΩ and 15 kΩ).

C. LEM HASS 50-S Current Sensor (U_{ref} = 2.5 V at 0 A, I_{PN} = 50 A):

$$I_{P(5kW)} = 5000\ W \div 300\ V = 16.67\ A \rightarrow V_{ref,cur} = 2.5\ V + 0.625\ V \times (16.67\ A \div 50\ A) = 2.708\ V \approx 2.71\ V$$

2. Sensor Datasheet Conversions and Threshold Equations

Threshold Net	Monitored Condition	Target V_{ref}	R_{top} / R_{bottom}	Divider Equation
$\mathbf{V_{ref\text{-}brake}}$	Hard Braking (1000 PSI on AiM Sensor)	**2.50 V**	R5 = 10 kΩ / R6 = 10 kΩ	5.0 V × (10 ÷ 20) = 2.500 V

$V_{ref\text{-}current}$	5 kW Motor Power (16.67 A at 300 V)	**2.71 V**	R7 = 11 kΩ / R8 = 13 kΩ	5.0 V × (13 ÷ 24) = 2.708 V
$V_{ref\text{-}4v8}$	Short-to-Power Fault (> 4.5 V Max Normal)	**4.80 V**	R1 = 1 kΩ / R3 = 24 kΩ	5.0 V × (24 ÷ 25) = 4.800 V
$V_{ref\text{-}0v2}$	Open-Wire / Ground Fault (< 0.5 V Min Normal)	**0.20 V**	R2 = 24 kΩ / R4 = 1 kΩ	5.0 V × (1 ÷ 25) = 0.200 V

3. Unity-gain Op-Amp Subtractor Architecture

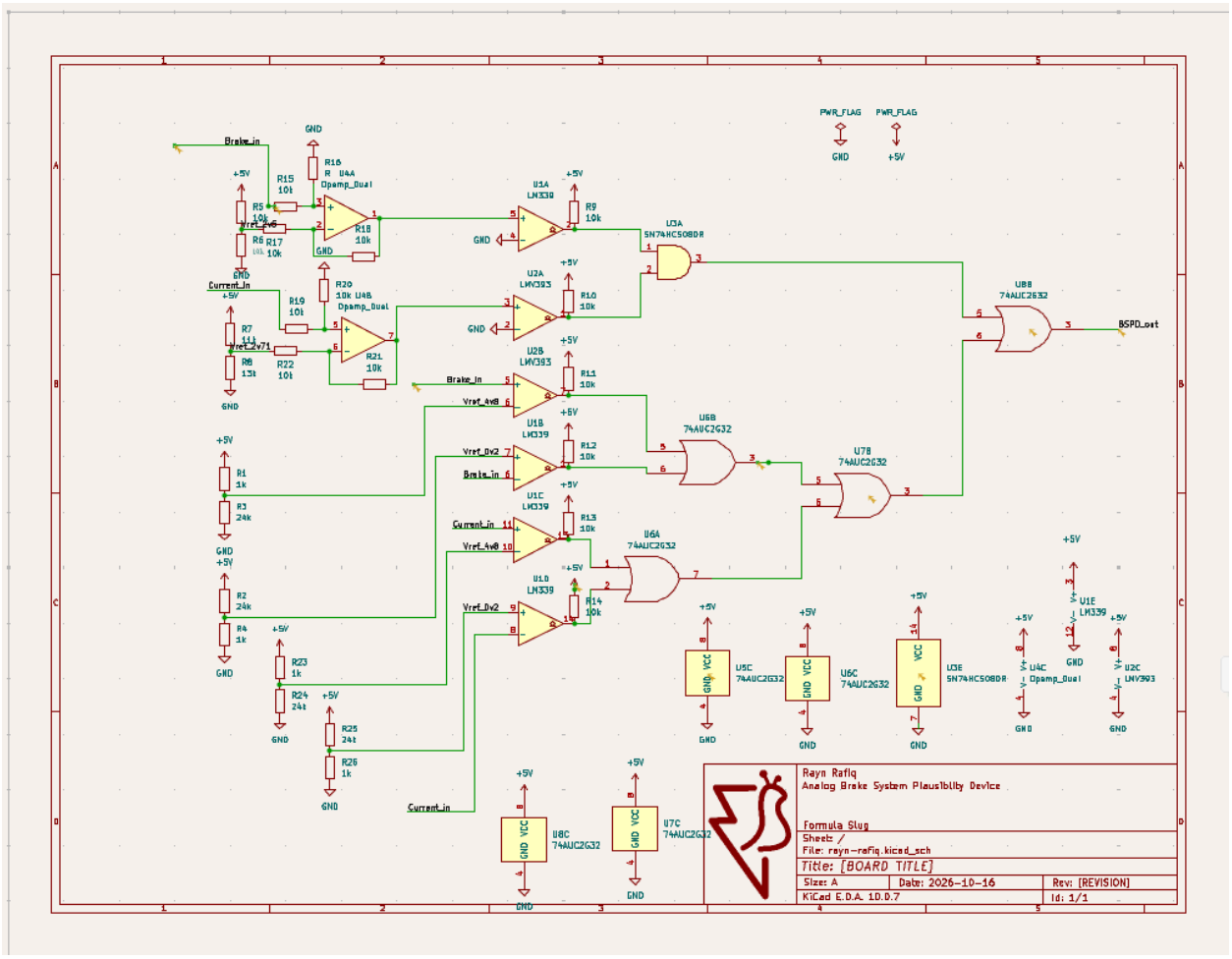
Rather than comparing two floating analog voltages directly on the main plausibility comparators, the circuit places a unity-gain **Difference Amplifier (Subtractor)** built from a dual op-amp (**U4A** for Brake, **U4B** for Current) and four matched **10 kΩ** resistors ahead of comparators **U1A** and **U2A**. Each op-amp computes $\mathbf{V_{diff}} = \mathbf{V_{sensor}} - \mathbf{V_{ref}}$. When the sensor voltage exceeds its threshold, $\mathbf{V_{diff}} > \mathbf{0\ V}$, which is compared directly against a ground **GND (0 V)** reference on the comparator's inverting (−) pin.

The four wire-fault comparators (**U2B, U1B, U1C, U1D**) do not require op-amp subtractors because broken wires or rail shorts force the signal all the way to 0 V or +5 V, allowing direct comparison against the fixed **4.80 V** and **0.20 V** dividers.

4. Comparators, Open-Collector Pullup Resistors and Logic Tree

- **Comparator Pin Polarities (U1A–U1D, U2A–U2B):** Short-to-power comparators (**U2B** for Brake, **U1C** for Current) place the sensor on the **+** pin and **4.80 V** on the **−** pin. Open-circuit comparators (**U1B** for Brake, **U1D** for Current) invert this arrangement by placing **0.20 V** on the **+** pin and the sensor on the **−** pin so that the output swings HIGH whenever the sensor drops below 0.20 V.
- **Open-Collector Pull-Up Resistors (R9–R14 = 10 kΩ):** Because the LM339 and LMV393 use open-collector output transistors that can only sink current to ground, a **10 kΩ** pull-up resistor to **+5 V** is wired to each of the six comparator outputs to generate a valid 5 V logic HIGH.
- **AND + Cascaded OR Gate Bracket & Relay K1:** The 2-input AND gate (**U3A**) asserts HIGH only when both U1A and U2A trip simultaneously. Four 2-input OR gates (**U6B, U6A, U7B, U8B**) form a tournament bracket combining the four wire-fault signals with U3A's output into `BSPD-out`, energizing Shutdown Relay(not shown)
- **IC Power Units & ERC Verification:** Dedicated power sub-units (**U1E, U2C, U3E, U6C, U4C**) connect all IC supply pins to **+5 V** and **GND**, with `PWR FLAG` symbols attached to both rails to pass KiCad Electrical Rules Check (ERC) with zero errors.

Final Schematic:


Brake_in
Current_in
Vref_4v8
Vref_2v71
Vref_0v2
BSPD_out
PWR_FLAG
GND
+5V
Rayn Rafiq
Analog Brake System Plausibility Device
Formula Slug
Sheet: /
File: rayn-rafiq.kicad_sch
Title: [BOARD TITLE]
Size: A
Date: 2026-10-16
Rev: [REVISION]
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